Chapter 7: Haloalkanes

PSPM 2011 / 2012

Q. 1 For the reaction of bromopropane, ***E*** below: 5m

$$\underset{\textbf{\textit{E}}}{CH_3CH_2CH_2Br} \xrightarrow{CH_3OK} \textbf{\textit{H}}$$

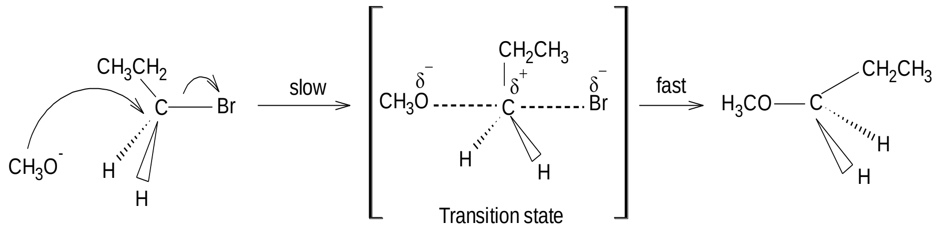
(a) Name the class of compound ***H***.
(b) Name the type of reaction that converts ***E*** to ***H***.
(c) Show the mechanism for the formation of ***H***.

Answer

(a) Ether

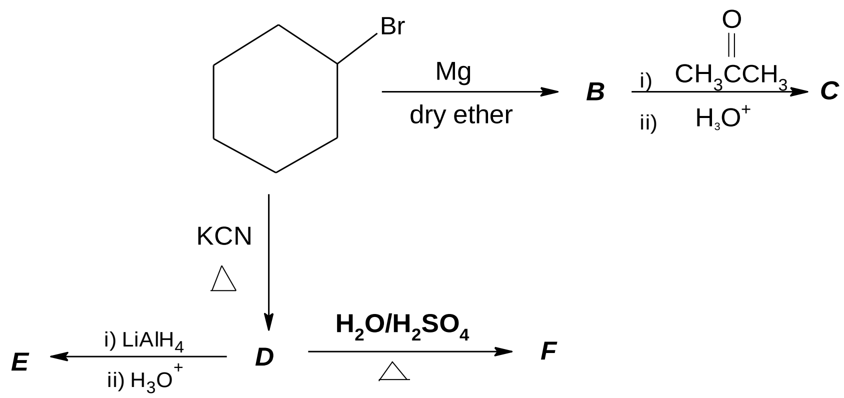
(b) S_N2 or nucleophilic substitution bimolecular

(c)



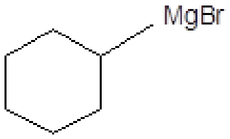
PSPM 2012 / 2013

Q. 2 (a) Draw the structure of the products ***B*** through ***F*** in the reaction below. 5 m

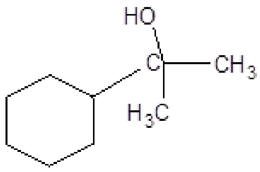


Answer

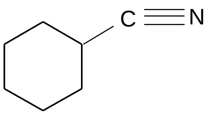
B =



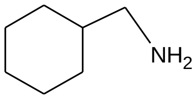
C =



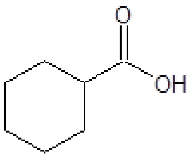
D =



E =



F =



Q. 2 (b) What does the symbol S_N stand for? Explain using an appropriate chemical equation, what you understand by an S_N reaction of an alkyl halide. 11 m

An alkyl halide ***V***, $C_5H_{11}Br$ undergoes hydrolysis to form an alcohol ***W***, $C_5H_{12}O$. alcohol **W** becomes cloudy immediately when reacted with a Lucas reagent. Deduce the structures of ***V*** and ***W***. State and propose the mechanism for the conversion of ***V*** to ***W.***

Answer (b) S_N = Nucleophilic substitution

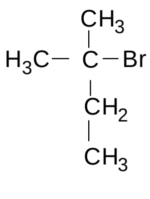
$RX + Nu^-$ $RNu + X^-$ OR give example

RX = any alkyl halide

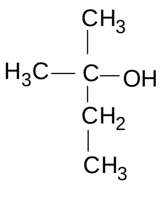
Nu^-= any nucleophile other than X

X is substituted by Nu^-

W = 3^0 alcohol



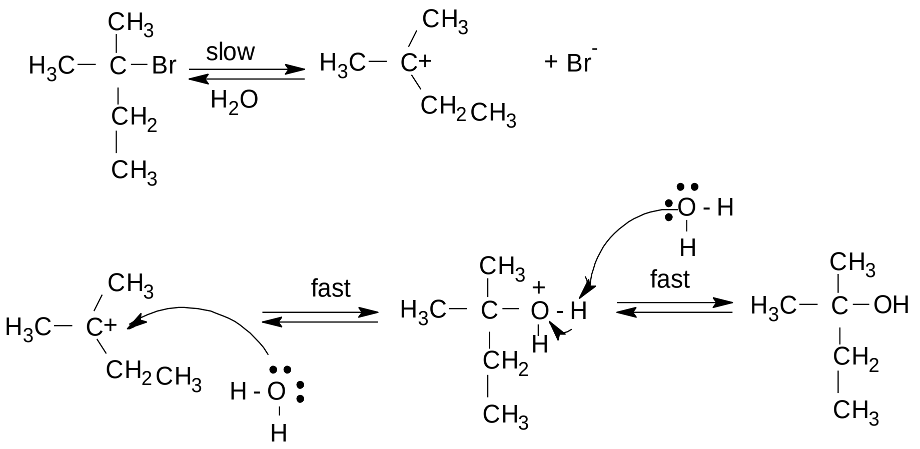
V



W

S_N1

Mechanisme



Remark:

If structure V is wrong, still can get mark max 2M as long as V is 3° alkyl halide

PSPM 2013 / 2014

Q. 3 (a) The reaction of an alkyl halide, ***X***, with sodium methoxide, $NaOCH_3$, produces 1-methoxybutane. 10 m

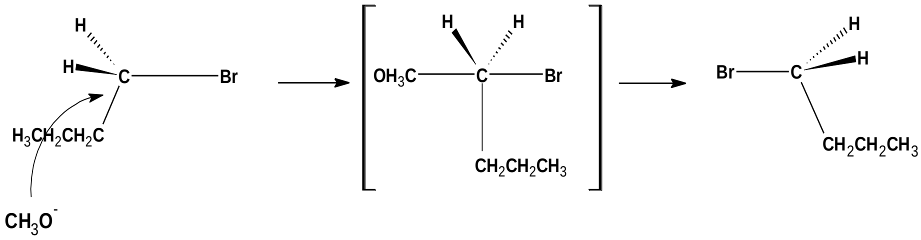
Using a suitable justification, propose the structure of ***X***. Write the mechanism for the formation of 1-methoxybutane. Name this mechanism.

If sodium methoxide is replaced with methanol, what is its effect on the rate of reaction above? Explain your answer. Give another factor that influences the rate of reaction.

Answer (a)

X = $CH_3CH_2CH_2CH_2Br$

Primary alkyl halide @ reaction occur at C1

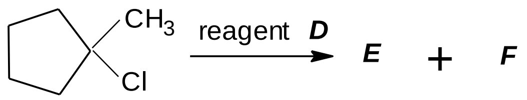


Type of reaction: Bimolecular nucleophilic substitution (S_N2).

The **reaction rate decreases** because CH_3OH is a **weak nucleophile**.

Concentration of R-X @ Nucleophile.

Q. 3 (b) Compound ***C*** reacts with reagent ***D*** to give unsaturated compounds, ***E*** and ***F***. 8 m



Compound ***C***

(i) Draw the structure of ***E*** and ***F***. Label the major product.

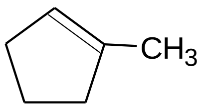
(ii) Explain how the major product is determined.

(iii) State reagent ***D***.

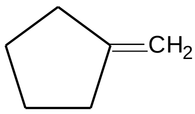
(iv) What is the function of ***D***? Explain your answer.

Answer (b)

(i)



E (major)



F *Note: Structure E and F are interchangeable

(ii) Saytzeff / Zaitsev Rule

Highly or more substituted alkene is more stable

(iii) D: C_2H_5ONa, C_2H_5OH,heat @ KOH, ethanol, Δ,

Note: OH^- or any RO^- can be used

(iv) Base

It removes proton from alkyl halide

PSPM 2014 / 2015

Q. 4 (a) **TABLE 1** shows the relative rates of S_N2 reactions for alkyl halides. 8 m

Class of Alkyl halide	Compound	Relative rate
Methyl	bromomethane	30
1°	bromoethane	1
2°	***A***	0.03
3°	2-bromo-2-methylpentane	***B***

TABLE 1

(i) Explain the relationship between the class of alkyl halide with the relative rates of reaction.

(ii) Give one example of compound A (a four carbon alkyl halide)

(iii) Predict the value of B. Give your reason.

(iv) Reaction of bromoethane with NaCN, NH3, and NaOCH3 gives C, D, and E respectively. Draw the structures for C, D, and E.

Answer (a) (i) Relative rate of S_N2 reaction is **higher for primary alkyl halide** (or less hindered) compared to tertiary alkyl halide (highly hindered) @ *vice versa*

(ii) 2-bromobutane @ 2-chlorobutane @ 2-iodobutane

(iii) Zero @ less than 0.03

(iv) Tetiary alkyl halide does not favor S_N2 due to bulkiness (**steric hindrance**)

C: CH3CH2CN
D: CH3CH2NH2
E: CH3CH2OCH3

Q. 4 (b) Arrange the following alkyl bromides in order of decreasing reactivity towards S_N1 reaction. Explain your answer. 10 m

2-bromopropane, 1-bromopropane, 2-bromo-2-methylpropane

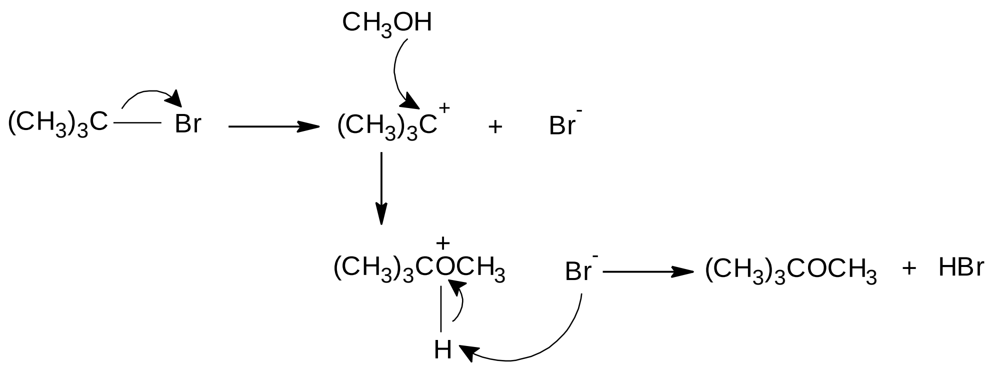
Write the chemical equation for the methanolysis of the most reactive alkyl bromide in S_N1 reaction above. Propose the mechanism for this reaction.

Answer (b) 2-bromo-2-methylpropane > 2-bromopropane > 1-bromopropane

2-bromo-2-methylpropane is a **tertiary alkyl halide** which will form the **most stable tertiary carbocation**.

$$(CH_3)_3Br \ + \ CH_3OH \xrightarrow[\Delta]{} (CH_3)_3COCH_3 \ + \ HBr$$

<u>Mechanism:</u>



PSPM 2015 / 2016

Q. 5 (a) Draw the structure of the product formed when 1-bromopropane reacts with the following reagents respectively: 4 m

(i) ammonia

(ii) methanol

(iii) sodium cyanide

(iv) water

Answer

(i) $CH_3CH_2CH_2NH_2$

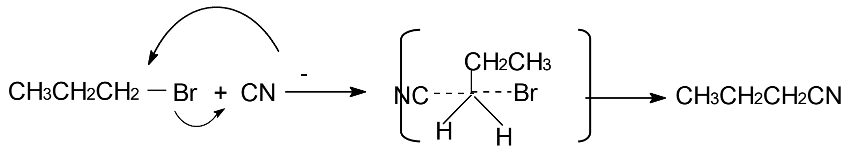
(ii) $CH_3CH_2CH_2-OCH_3$

(iii) $CH_3CH_2CH_2CN$

(iv) $CH_3CH_2CH_2OH$

Q. 5 (b) Draw and name the mechanism for the reaction between 1-bromopropane with sodium cyanide. 4 m

Answer (b) **Mechanism : S_N2**



Q. 5 (c) Pentanoic acid can be prepared from 1-butanol, F , as shown below: 7 m

$$\underset{\mathbf{F}}{\mathbf{CH_3CH_2CH_2CH_2OH}} \xrightarrow{PCl_5} \mathbf{G} \xrightarrow[\text{Dry ether}]{\text{Mg}} \mathbf{H} \xrightarrow[ii.\ Y]{i.\ X} \mathbf{CH_3CH_2CH_2CH_2COOH}$$

(i) Draw the structures of G and H.

(ii) State reagents X and Y.

(iii) Give the reagent needed to prepare F from 1-butene.

(iv) Suggest an alternative two-step reaction to convert G to pentanoic acid.

Answer

(c) (i) G : $CH_3CH_2CH_2CH_2Cl$

H : $CH_3CH_2CH_2CH_2MgCl$

(ii) X : CO_2

Y : H_3O^+ @ H_2O /H^+ H_2O,H^+

(iii) (i) HBr , H_2O_2 , (ii) NaOH

(iv) $\mathbf{G} \xrightarrow{CN^-} CH_3CH_2CH_2CH_2CN \xrightarrow{H_3O^+} CH_3CH_2CH_2CH_2COOH$

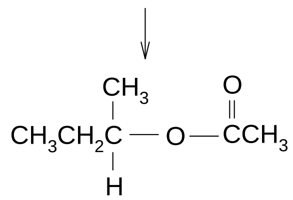
@ KCN

@ NaCN

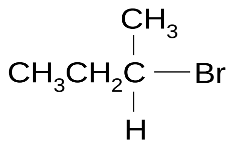
Q. 5 (d) An alkyl bromide EE reacts with CH3CH2OH to give FF, whereas compound $CH_3COOCH(CH_3)CH_2CH_3$ is formed when EE is treated with a nucleophile.Dehydrohalogenation of EE produces GG which obeys Saytzeff`s rule.
Deduce the structures of EE, FF, GG and nucleophile. Write reaction equation for all reactions involved. 10 m

Answer

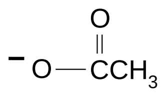
alkyl bromide, *EE* + nucleophile



EE : 2° alkyl halide :



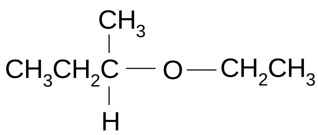
nucleophile



:

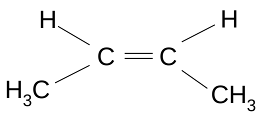
FF is an ether yield from the reaction between alkyl halide and alcohol.

FF :

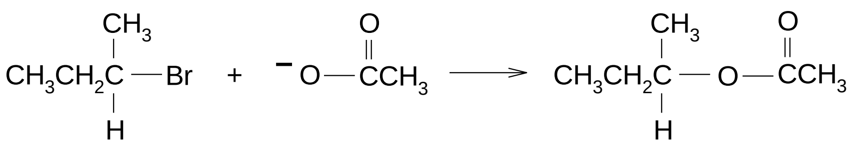


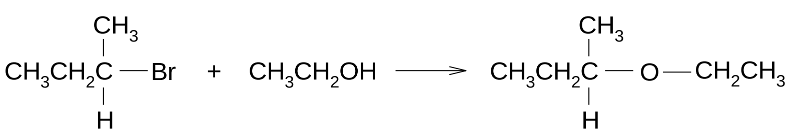
Saytzeff'srule : more highly substituted product.

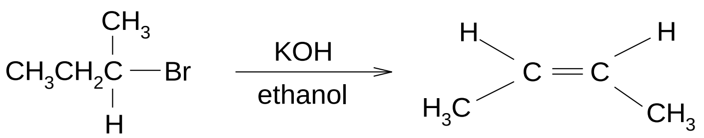
GG :



Reaction equation :



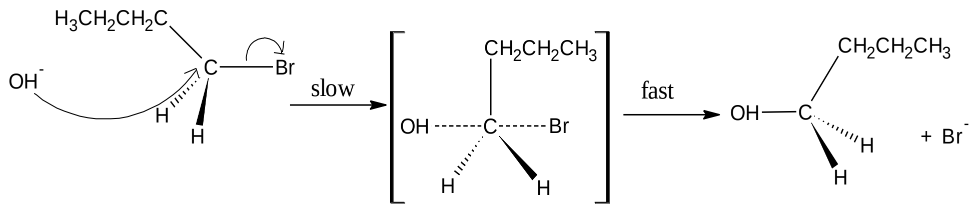




PSPM 2016 / 2017

Q. 6 (a) State and show the mechanism for the reaction of 1-bromobutane with the hydroxide ion. 4 m

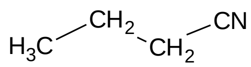
Answer (a) Mechanism : S_N2



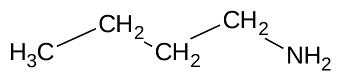
Q. 6 (b) 1-chloropropane reacts with KCN in ethanol under reflux to form **HH**. Reduction of **HH** produce **JJ** while hydrolysis of **HH** using aqueous acid, under reflux gives **KK**. Grignard reagent of 1-cholropropane reacts with ethanal to form **LL**. Show the formation of this Grignard reagent. Draw the strucyures of **HH**, **JJ**, **KK**, **LL** and write the chemical equation for the formation of **JJ** and **LL**. 5 m

Answer

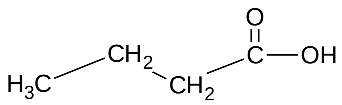
HH



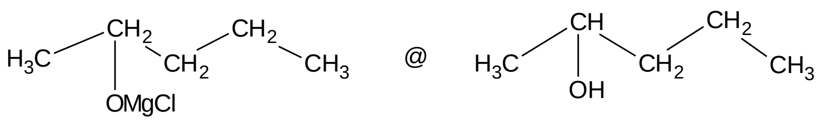
JJ

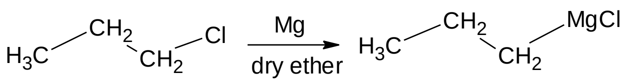


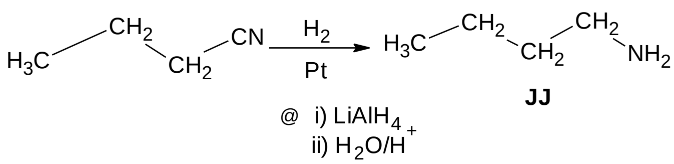
KK

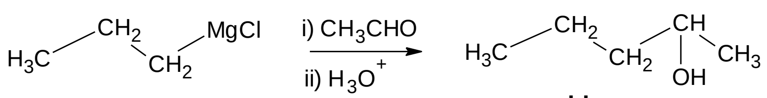


LL









LL

PSPM 2017 / 2018

Q. 7 (a) Treatment of 2-bromo-2-methylbutane with water yields compound E. The dehydration of E gives compounds F and G. 10 m

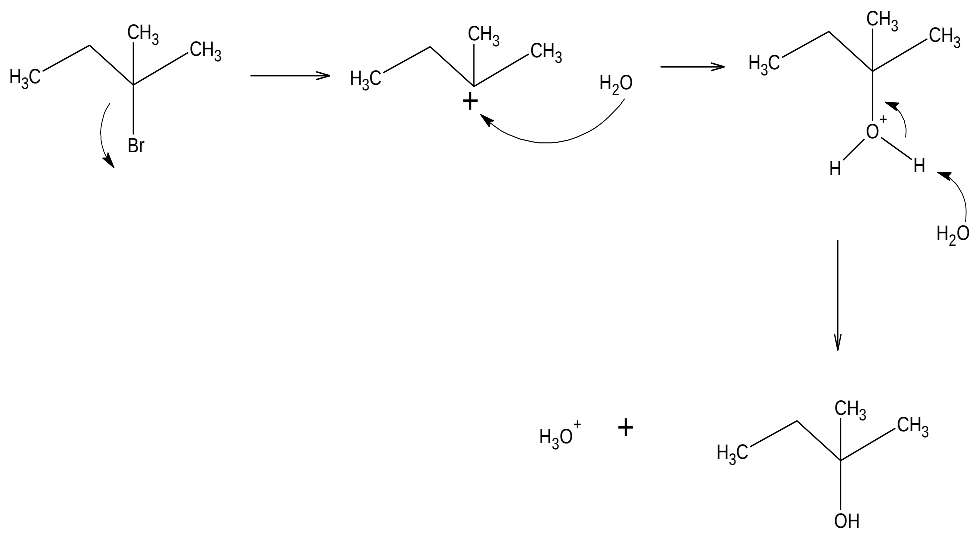
(i) State the reagent and reaction condition used in the dehydration of E.
(ii) Draw the structural formulae of compounds E, F and G.
(iii) Indicate the major product formed in the dehydration reaction. Explain your answer.
(iv) Draw the mechanism for the formation of E.

Answer (a) (i) conc. H_2SO_4, heat/reflux/Δ @ H_3PO_4, heat/reflux/Δ

(ii) E = $CH_3CH_2C(CH_3)(OH)CH_3$
F/G = CH_3=$CHC(CH_3)_2$
G/F = CH_2=$C(CH_3)CH_2CH_3$

(iii) CH_3CH=$C(CH_3)_2$
Follow Saytzeff's rule

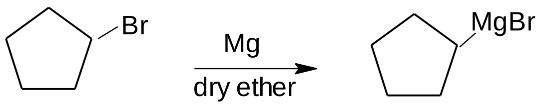
(iv)



Q. 7 (b) Grignard reagent is a versatile tool in synthetic organic chemistry. Using bromocyclopentane as a starting material, show how a Grignard reagent, **X**, is synthesized. 10 m

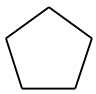
(i) (ii) (iii) Reaction of **X** with water produces compound **Y** while treatment in carbon dioxide followed by hydrolysis forms compound **Z**. 3-methyl-2-butanone reacts with **X** and hydrolyses to yield compound **AA**. Draw the structural formulae of compounds **Y**, **Z** and **AA** and write the chemical equations respectively.

Answer (b)

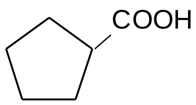


X

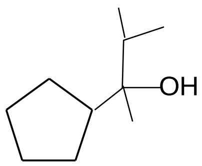
(Grignard reagent)



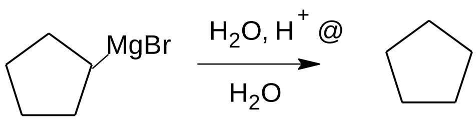
Y



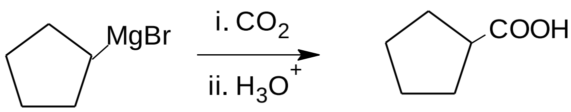
Z



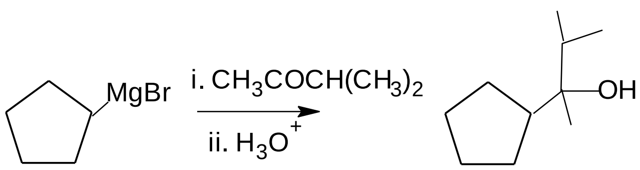
AA



Y



Z



AA

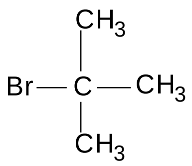
PSPM 2018 / 2019

Q.8 Two organic compounds **D** and **F** with molecular formula C_4H_9Br react with suitable reagents to form **E** and **G** *via* S_N1 and S_N2 reactions respectively. Both **E** and **G** have a molecular formula of $C_4H_{10}O$. 8m

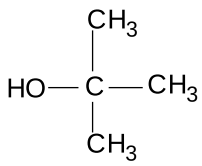
(a) Draw the structural formulae for **D, E, F and G.**

Answer (a)

D :



E :



F : $CH_3CH_2CH_2CH_2Br$ @ $(CH_3)_2CHCH_2Br$

G : $CH_3CH_2CH_2CH_2OH$ @ $(CH_3)_2CHCH_2OH$

(b) Suggest reagents for the formation of **E** and **G**.

Answer (b)

E : H_2O

G : NaOH @ KOH

(c) Write the mechanism for the formation of **E.**

Answer (c)